

MELATONIN

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Melatonin is a molecule that has been found in every animal and plant studied to date, from human beings to the most primitive one-celled algae that evolved more than three billion years ago. In each organism, the molecular structure of melatonin is identical. This sameness is a rare occurrence in nature. In all the life forms studied, melatonin has been produced in the same circadian (daily secretion pattern) rhythm, with higher levels produced at night than during the daytime.

In humans, melatonin is produced and secreted by the pineal gland in the brain. It is synthesized from the brain chemical (neurotransmitter) serotonin, which itself is synthesized from 5-hydroxytryptophan, which is originally derived from the amino acid tryptophan, found in many foods. In the human body melatonin acts as a hormone, neurotransmitter (brain chemical), antioxidant and immune system stimulator. This tremendous versatility accounts for the many profound effects that melatonin exerts on our health status.

As a neurotransmitter (brain chemical) melatonin is best known for its ability to induce sleep in persons with insomnia and to remedy problems with interrupted sleep, as well as its ability to elevate mood. Humans produce five to ten times more melatonin at night than during the day, a circadian rhythm found in animals as well. Peak amounts occur around two or three o'clock in the morning. The darker the room, the more melatonin is secreted by the pineal gland at night.

After puberty the body's production of melatonin has been shown to decline. By age 40 humans often produce approximately 60 percent less melatonin than a 10 year old and by age 70 or 80 melatonin levels may be undetectable. The use of certain drugs, such as aspirin, ibuprofen, and

beta-blockers (used to control high blood pressure) also contributes to a decline in melatonin levels.

With respect to its antioxidant properties, melatonin is unique in that it is both a water-soluble antioxidant and a fat-soluble antioxidant. This means that it can quench and intercept dangerous free radicals in every area of the body and in every part of every cell in the body. Not even vitamin C or vitamin E has this range of antioxidant capability. Some studies suggest that as an antioxidant melatonin is twice as effective as Vitamin E, five times as efficient as glutathione, and five hundred times more effective than the synthetic DMSO (dimethyl sulfoxide). However, melatonin is produced in picograms (a trillionth of a gram – the smallest amount of any hormone), whereas Vitamin E, and other antioxidants, are present in the body at much higher concentrations.

The decline in melatonin synthesis and secretion by the pineal gland is considered to be a very significant factor that contributes to less restful sleep, accelerated aging and increased risk of certain cancers, dementia, Alzheimer's disease, and a weakening of our immune system as we age. Many anti-aging experts recommend taking a melatonin supplement by age 50, as studies suggest that providing more youthful levels of melatonin to your brain and body can help combat aging, boost immune system function, reduce cancer and heart disease risk and produce better quality sleep and mood enhancement.

Melatonin for a Good Night's Sleep

We sleep roughly 16 hours as infants, 8 hours in adolescence, 7 hours as adults, and less than 6 hours in our geriatric years. No one seems to be exactly sure how melatonin induces better quality, restful sleep, but the reduced output of melatonin as we age is strongly associated with decreased sleep duration, a greater tendency towards interrupted sleep, and a lesser ability to achieve deeper levels of sleep. Delta sleep, the deepest most restorative stage of sleep is longer in

childhood, decreases at puberty, and progressively declines after age 30. The elderly spend less time in this restorative stage and therefore, are likely to feel tired during the day. This accounts for the fact that as we get older we often feel the need to rest more, but rarely feel completely rested, rejuvenated, and completely re-energized after a night's sleep, as we did when we were children.

Melatonin is made in the pineal gland from serotonin, which is a mood enhancing, appetite suppressing, brain chemical (neurotransmitter). Daylight stimulates the production of serotonin in our brain. That is one of the reason we feel so much better on a bright, sunny day, as the more sunlight that enters the pupil of the eye, the greater is the synthesis of serotonin in our brain. Darkness allows serotonin to be converted into melatonin in the pineal gland, and the darker the room the more melatonin we secrete through the night. Starting in the evening, and through out the night, the pineal gland releases melatonin, reaching peak levels between 2 and 4 am. In the morning, exposure to light shuts off melatonin production.

An interesting study reported in the British Medical Journal in 1994 showed that the quality of sleep in elderly individuals is proportional to the amount of melatonin secreted by their pineal gland. Evidence suggests that cells within the pineal gland decrease in number as we age, making them less efficient in producing melatonin, which in part, accounts for the decline in melatonin secretions as we get older. Studies show that melatonin supplementation taken one hour prior to bedtime helps individuals proceed more quickly into stage 2 sleep (early stage of sleep) and remain longer in the deeper levels of sleep (stage 3, 4, and REM sleep-dream time sleep) than occurs without melatonin supplementation.

One of the ways that melatonin supplementation can strengthen our immune system is by helping us attain deeper levels of sleep, as it is known that the immune system is most active during stage 3 and stage 4 sleep (delta sleep). However, many immune cells are also directly influenced by melatonin, which has been shown to enhance many parameters of immune function.

Studies conducted on humans have shown that melatonin is an effective sleep aid, allowing older subjects to experience the deeper, rejuvenating, immune-enhancing form of sleep we enjoyed in our youth. As a sleep aid, melatonin has been shown to be safer than other sleeping pills, which come with a host of side effects and toxicity problems. Melatonin toxicity studies demonstrate that it is not only extremely non-toxic and safe, but it also provides many other health-promoting, anti-aging benefits to the body beyond improving our sleep quality. Taken at the right dosage, one hour before bedtime, melatonin does not produce morning drowsiness, and helps to elevate mood and feeling of well being though out the day, which is probably due to its chemical resemblance to serotonin – the mood elevating brain chemical. Anti-aging experts suggest that we begin using melatonin supplementation between age 40 and 50, as a means to improve sleep quality, support immune function, and to help reduce our risk of cancer, dementia, Alzheimer's disease and possibly heart disease.

Melatonin May Prevent Breast and Prostate Cancer

The decline in melatonin secretion as we age may allow breast cancer and prostate cancer to develop more easily, which may partially account for the increased incidence in these cancers as we get older. Studies demonstrate that melatonin has an anti-estrogen effect on reproductive tissues, which has been shown to tone down the over stimulation effect of estrogen on breast

cells, an effect that is associated with decreased risk of breast cancer development. Some early studies demonstrated that women who sleep in darker rooms throughout their lifetime, and who therefore have higher secretion rates of melatonin through the night, have a lower risk of breast cancer during their lifetime than do women who sleep in a room that is less dark.

More recently, experimental studies have shown that melatonin can reduce the replication rate of human breast cancer cells by 50-75%, when these cells were exposed to alternating levels of melatonin to mimic the body's day and night time melatonin secretion patterns. This was only true in regards to breast cancer cells that possess estrogen receptors on their surface (estrogen-receptor-positive breast cells), which indicates that melatonin likely latches on to estrogen receptors, as does the hormone estrogen, but rather than stimulate these cells to divide rapidly, it inhibits or slows their cell division rate. This is similar to the action of the drug Tamoxifen, which is used to prevent the recurrence of breast cancer and its metastases, in women that have estrogen-receptor-positive breast cancer. In fact, experimental studies have shown that melatonin can enhance the ability of Tamoxifen to inhibit the replication of breast cancer cells. These studies prompted Dr. Paolo Lissoni to combine Tamoxifen therapy and melatonin supplementation (20 mg twice per day) in a trial of breast cancer patients. The study showed that four out of the fourteen women had a 50% or greater reduction in the size of their tumors. Eight additional women showed no further progression in tumor size. Only two women failed to respond to the treatment. These results were better than the typical outcomes associated with Tamoxifen use alone. Larger studies of this kind are underway presently to better understand the use of melatonin as a cancer therapy agent.

However, from the standpoint of prevention, the body of evidence suggests that melatonin may protect against breast cancer in several ways. First, it antagonizes the effects of estrogen on breast cells, helping to slow down the rate of cell division, which in turn, decreases the chance of a cancer cell developing. Second, should a cancer cell develop in the breast, melatonin has been shown to inhibit its ability to divide and spread, giving the body's immune system a better opportunity to kill it off before it can really take hold and become a life threatening problem. Thirdly, melatonin enhances the effectiveness of the Natural Killer Cells, an important immune cell that seeks out developing cancer cells and kills them before they can start to divide and thrive in the body. Also keep in mind that melatonin is a powerful antioxidant that helps to prevent free radical damage (and the genetic mutations they cause) to all body tissues and all cellular components, including the DNA of breast cells. Thus, the lower melatonin levels that are found in a woman's body after 50 years of age (men and women 51-55 years of age secrete half as much melatonin at night as is secreted at age 21-25; melatonin secretion is four times lower in men and women 82-86 years of age than at age 21-25) may reduce a woman's ability to prevent the development and spread of breast cancer. Thus, many anti-aging experts recommend that women take 200 mcg (.2 mg) -3,000 mcg (3 mg) of melatonin an hour before bedtime each night, beginning at age 40-50, to re-acquire the protective levels of melatonin that may help guard against breast cancer in younger women.

In regards to prostate cancer, studies by R. Philo and A. Berkowitz, published in the Journal of Urology in 1988, showed that melatonin inhibits the growth and replication of human prostate cancer cells. More recently, an experimental study by David Blask and another group of researchers demonstrated similar findings, showing a 50% reduction in prostate cancer cell growth and replication when these cells were incubated with melatonin. Like breast cancer

patients, males with diagnosed prostate cancer have been shown to have low blood levels of melatonin. However, we are not sure at this point if the low melatonin levels increased the risk of prostate and breast cancer development or if these cancers cause a reduction in melatonin levels.

There is one important case study that showed that melatonin supplementation at 6 mg per day was able to reduce the PSA level from 30 to 4 within six months in a confirmed case of localized prostate cancer. The PSA (prostate specific antigen blood level) is an indirect marker of the activity of the tumor. A normal PSA level is 0-4. Thus, it may be advantageous for men with prostate cancer, who are in the watchful waiting stage of the condition, to take 6-10 mg of melatonin one hour before bedtime each night. This intervention may help reverse or better control prostate cancer cell activity, which may help improve outcomes from concurrent or future medical interventions (drugs, surgery, radiation treatment).

However, in regards to prevention, the body of evidence suggests that melatonin supplementation by age 40-50 years of age, may be another intervention that can help deter development of prostate cancer, which remains the most frequently diagnosed cancer in men in North America.

Dr. Lissoni and colleagues, from the division of oncological radiotherapy at Gerardo Hospital in Milan, Italy, have been on the forefront of melatonin research as it applies to cancer prevention and treatment. They have administered melatonin in daily doses of 20 mg per day, in conjunction with the use of standard drugs and related cancer treatments, in cases of advanced lung, colon, and metastatic brain cancer. Their preliminary studies have shown that melatonin can improve

outcomes in a significant number of these cases, when added to traditional medical treatment. Further studies are underway to evaluate the role of melatonin as an adjunct to cancer treatment for a variety of cancers. In the meantime, its ability to inhibit the growth of human breast and prostate cancer cells under experimental conditions, along with its other anti-cancer properties, make melatonin supplementation a strong consideration for men and women over 40 years of age, to compensate for the age-related decline in melatonin secretion rates that become significant at this stage in life.

Melatonin Boosts Immune Function

Many studies have shown that melatonin can boost various aspects of immune function and thus, the decline in melatonin levels as we age may be an significant contributing factor to the parallel decline in immune function that occurs after age 40 to 50. Receptors for melatonin have been found in lymphoid organs such as the thymus gland and spleen, and on certain white blood cells. Some studies suggest that melatonin can help restore thymus gland function and prevent age-related thymus gland degeneration. Studies on mice demonstrate that melatonin can enhance immune function even when mice were given cortisol, a drug known to suppress the immune system. Melatonin supplementation also showed that it could keep the immune system working at a more optimal level in mice that were subjected to various types of stress. Stress is also known to suppress immune function, but melatonin administration buffered the immune system against the effects of stress to a significant degree in these studies.

The work of researchers G.J. Maestroni and A. Contri, and fellow investigators, has provided us with an important piece of the puzzle as to how melatonin helps to optimize the function of our immune system. They have shown that the T-Helper Cells of our immune system have receptors for melatonin. When melatonin attaches to these receptors a cascade of events is set into motion that significantly boosts our immune function. Once melatonin docks to the receptors on our T-Helper Cells, they in turn stimulate the release of an important immune system signaling agent known as interleukin-4 (IL-4), which increases the production and efficiency of Natural Killer Cells, and stimulates immune cells to more efficiently engulf and destroy various micro-organisms (viruses, bacteria, yeast fungi etc.). Studies show that taking 2 mg of melatonin at night can increase the production of Natural Killer Cells by 240% compared to pre-melatonin supplementation levels. Natural Killer Cells specialize in attacking cancer cells and virus-infected cells, two extremely important aspects in the prevention of cancer and the infections disease process, served by our immune system. The release of IL-4 also stimulates the production of a hormone-like substance (granulocyte-macrophage colony-stimulating factor) that increases the production of white blood cells in our bone marrow. White blood cells are the cells of the immune system that battle infections and cancer cells in our bloodstream and in all the tissues and organs of the body.

The importance of T-Helper Cells in coordinating the movements, proliferation and efficiency of our immune cells (white blood cells) is highlighted by the fact that the primary way in which the HIV-virus exerts its lethal effects is by infecting our T-Helper cells, suppressing their function and decreasing their numbers in the body. When T-Helper Cell concentrations fall below a critical threshold (200 cells per microliter) due to infection with the HIV-virus, this is the point where AIDS begins, whereby opportunistic infections, general deterioration and increased

susceptibility to malignancy become more prevalent in these cases. In fact, studies by G. Maestoni and fellow researchers have shown that melatonin supplementation of 20 mg per day (taken at 7:00 PM), given to fourteen patients with HIV infection or full blown AIDS was able to raise the T-Helper Cell blood count by 35%, on average. It also raised the concentrations of Natural Killer Cells by 57%, and raised the levels of one type of lymphocyte by 76%. None of the patients had any adverse side effects from melatonin supplementation at this dosage. Other studies by Dr. Maestoni as well as Dr. Lissoni, have shown that the melatonin supplementation can produce a number of favorable outcomes on the immune system in HIV and AIDS patients, when administered in conjunction with other standard HIV drugs. These studies provide definitive evidence that melatonin enhances important aspects of immune function, even in individuals with compromised immune diseases.

There is reason to believe that daily melatonin supplementation after age 40-50, when our body's secretion rates of this hormone have reached a critical point of decline, may be highly advantageous in preventing the deterioration in immune function that commonly accompanies the normal aging process and makes us more prone to cancer and life-threatening infectious diseases.

Melatonin May Reduce Risk of Alzheimer's Disease and Age-Related Cognitive Impairment

Melatonin is not only made by a gland in the brain (pineal gland), but may be the brain's most powerful antioxidant agent helping to prevent free radical damage to our brain cells throughout our lifetime. The brain uses at least 10% of the body's oxygen at given moment, which it

requires to produce the energy brain cells need to for us to think and for the brain to process all the conscious and unconscious information it responds to each moment of our lives. The brain is a very metabolically active tissue that consumes much of the body's oxygen supply and hence, generates an abundant amount of oxygen free radicals. Studies reveal that as much as 5% of the oxygen consumed by our tissues create free radicals as a by-product of oxygen metabolism. As the brain uses a great deal of the body's oxygen, it also generates a great deal of oxygen free radicals in the process, and these free radicals are known to damage brain cells in such a way as to contribute to the development of Alzheimer's disease, age-related decline in cognitive function, and is associated with other neurodegenerative diseases (multiple sclerosis, Parkinson's disease, Lou Gehrig's disease, etc.)

One study shows that almost half of people over age 85 have Alzheimer's disease. In their quest to understand which elderly individuals are most at risk for the development of memory loss and/or Alzheimer's disease, Italian gerontologists discovered that older individuals with higher melatonin levels had better mental acuity than those with low melatonin levels based upon a test known as the Mini Mental State Exam. Japanese researchers have revealed the same finding in that healthy functioning elderly subjects produce twice as much melatonin as Alzheimer's disease patients of similar age, according to their investigative study. Researchers argue that melatonin quenches oxygen free radicals produced in the brain sparing brain cells from their damaging effects.

Free radicals have been shown to be a significant factor in brain cell damage characteristic of Alzheimer's-type dementia, as evidenced by the fact that the brains of Alzheimer's patients contain higher levels of amyloid protein, which is not only an end product of free radical attack,

but also generates additional free radicals within brain cells itself. Melatonin has been shown to block the production of amyloid protein in animal studies, and is the antioxidant that most easily gains entry into the brain. Many experts believe that the decline in melatonin secretion as we age weakens the brain's antioxidant protection against oxygen free radicals, allowing them to more easily damage brain cells, promoting memory loss, decline in cognitive function, the development of Alzheimer's disease and contributes to the development or progression of other neurodegenerative conditions. Melatonin also increases the synthesis of glutathione peroxidase, which is another antioxidant (antioxidant enzyme) that has been shown to have a powerful effect on preventing free radical damage in the brain.

A large intervention study in recent years (Alzheimer's Disease Cooperative Study), demonstrated that high doses of Vitamin E (2,000 IU per day) could slow the progression of moderately advanced Alzheimer's disease in human subjects. This effect was thought to be mediated through the antioxidant function of Vitamin E. However, as mentioned earlier melatonin more easily enters the brain than any other antioxidant nutrient and has also been shown to quench free radicals in brain cells and boost brain levels of the antioxidant enzyme glutathione peroxidase. As such, researchers have begun work to test whether melatonin supplementation can prevent Alzheimer's-type conditions in animals.

Thus far, studies have revealed that melatonin can block important steps in the development of Alzheimer's disease in animal testing. In a landmark study melatonin has recently been shown to reverse the protein complex that is the hallmark of Alzheimer's disease, according to a published report in the December 11, 2001 issue of the journal, *Biochemistry*, published by the American Chemical Society.

The findings of the study were based upon experiments with animals and human brain cell cultures. The researchers conclude, “Our results clearly demonstrate the ability of melatonin to inhibit the process of forming the signature amyloid protein bundles seen in Alzheimer’s disease”. In Alzheimer’s disease, toxic fibrillar aggregates of a protein called amyloid beta protein (amyloid protein) are the pathologic hallmark of the disease. In this study melatonin inhibited the formation of amyloid protein, which is toxic to nerve cells and accelerates free radical damage to neurons.

In this study, only melatonin protected brain cells from amyloid beta protein build up and toxicity. Trials with Vitamin C, Vitamin E and a synthetic antioxidant were not protective in this experimental study.

More research is required before any definitive statements can be made regarding the use of melatonin and the prevention of Alzheimer’s disease; however, it appears that the prevention of excessive damage to our brain cells by oxygen free radicals may be another important reason to consider the use of melatonin supplementation, begin between age 40 and 50.

Melatonin Supplementation in Anti-Aging and Disease Prevention

Replacing the melatonin that the body no longer produces as we age may be an important aspect of anti-aging and disease prevention. Melatonin is a very powerful antioxidant that quenches free radicals in all body cells and tissues that are known to accelerate aging and increase our risk of cancer. Melatonin supplementation has been shown to improve sleep quality and enhance immune function, reversing certain aspects of the age-related decline in immune function. Experimental and observational studies suggest that melatonin supplementation may help reduce risk of breast and prostate cancer, and possibly other cancers. Melatonin is a powerful brain antioxidant that protects the brain from free radicals linked to Alzheimer disease, age-related cognitive decline and other degenerative conditions of the nervous system. Due to the significant decline in melatonin secretion that occurs between age 40 and 50, many experts suggest taking a melatonin supplement each day, beginning at this stage in life, to re-establish more youthful levels of this important hormone and capitalize upon the many anti-aging and disease prevention benefits it has been shown to provide.

Dosage Ranges of Melatonin

For anti-aging/disease prevention and/or sleep quality improving purposes, a nighttime dosage of 200 mcg (.2 mg) to 3,000 mcg (3 mg) is typically used. It is best to start with a small dose, in the range of 200-300 mcg, and work up, based upon improved sleep with no awakening through the night or morning drowsiness. Taking doses that are too high can result in morning grogginess, headaches and/or abdominal cramps, and vivid dreams that may awaken you through the night.

Taking therapeutic doses of melatonin as an adjunct to the treatment of a disease or condition, in the range of 5 – 75 mg per day, as is often used, should only be undertaken with knowledge of a physician, who is able to monitor blood work and other parameters, as these dosages are still experimental and long-term safety is unknown at this time

However, melatonin testing in humans and animals indicate that it is very non-toxic.

Contraindications and Drug-Nutrient Interactions

Some cases of depression can be aggravated by melatonin supplementation. Some antidepressant drugs stimulate the production of melatonin and others, such as fluoxetine (Prozac), may lower melatonin levels. It is probably best at this time not to combine melatonin with any other drug that targets brain chemistry or neurotransmitter levels (e.g. antidepressants, and drugs used to manage bipolar disease, schizophrenia etc.

A number of drugs have been shown to deplete melatonin levels, which may induce a need to begin replenishing melatonin at an earlier stage in life by those taking these drugs, which include, aspirin, ibuprofen, beta-blockers, and calcium channel blockers.

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